

Parishram 2027
Biology
Principles Of Inheritance and Variation

DPP: 4

Q1 Which of the following phenotypic ratios most likely indicates non-Mendelian interaction (epistasis) in a dihybrid cross?

- (A) 9 : 3 : 3 : 1 (B) 1 : 2 : 1
(C) 12 : 3 : 1 (D) 3 : 1

Q2 Statement I: In polygenic inheritance, the F_2 phenotypic ratio follows a bell-shaped curve. Statement II: The bell-shaped curve arises because each gene contributes equally and additively to the phenotype

- (A) Both statements are true, and II explains I
(B) Both are true, but II does not explain I
(C) I is true, II is false
(D) Both statements are false

Q3 A gene A is completely linked with gene B. If $AaBb \times aabb$ is crossed, what is the phenotypic ratio in the offspring assuming no crossing over?

- (A) 1 : 1 (B) 3 : 1
(C) 9 : 3 : 3 : 1 (D) 1 : 2 : 1

Q4 Mendel's Law of Independent Assortment fails when two genes are located on the same chromosome and are ____; in such cases, they are said to be ____.

- (A) close together; linked
(B) far apart; epistatic
(C) at centromeres; recombined
(D) on homologous chromosomes; separated

Q5 Match the Following

Column I

A) 9 : 3 : 3 : 1

B) 1 : 2 : 1

Column II

I) Complete independent assortment

II) Co-dominance

C) Continuous variation

D) 1 : 1

III) Polygenic trait

IV) Test cross of linked genes

(A) A-I, B-II, C-III, D-IV

(B) A-III, B-IV, C-II, D-I

(C) A-I, B-IV, C-II, D-III

(D) A-II, B-I, C-IV, D-III

Q6 A researcher crosses two pure lines of wheat differing in skin color. F_1 shows intermediate color. In F_2 , she observes five different phenotypic classes. How many gene pairs are likely involved in this trait?

- (A) 1 (B) 2
(C) 3 (D) 4

Q7 Two genes are located 10 map units apart on the same chromosome. What proportion of gametes produced by a heterozygote will be recombinant?

- (A) 5% (B) 10%
(C) 20% (D) 90%

Q8 Identify the incorrectly matched pair regarding inheritance patterns

- (A) Skin color – Polygenic
(B) Pea seed shape – Monohybrid
(C) Height in humans – Multiple alleles
(D) Eye color – Polygenic

Q9 Assertion (A): Mendel could not explain the chromosomal basis of inheritance.

Reason (R): Mendel was unaware of the existence of chromosomes and DNA during his time.

- (A) Both A and R are true, and R is the correct explanation



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- (B) Both A and R are true, but R is not the correct explanation
(C) A is true, R is false
(D) A is false, R is true

Q10 Which of the following are true regarding linked genes?

- I. They do not follow Mendel's law of independent assortment.
II. They always segregate together during

gamete formation.

III. Crossing over between them leads to recombination.

IV. Linkage strength is inversely proportional to the distance between genes.

(A) I, III, and IV

(B) I and II only

(C) All four

(D) II, III, and IV



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Answer Key

Q1 C
Q2 A
Q3 A
Q4 A
Q5 A

Q6 B
Q7 B
Q8 C
Q9 A
Q10 A



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Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

A 12:3:1 ratio suggests dominant epistasis, a deviation from the typical 9:3:3:1 ratio.

Video Solution:



Q2 Text Solution:

Additive effect of alleles in polygenic inheritance results in a normal distribution curve.

Video Solution:



Q3 Text Solution:

Linked genes inherited together give only two combinations: AB and ab. Offspring: AaBb and aabb 1:1 ratio.

Video Solution:



Q4 Text Solution:

Close proximity prevents independent assortment and causes gene linkage.

Video Solution:



Q5 Text Solution:

9:3:3:1 dihybrid; 1:2:1 co-dominance; continuous variation polygenic; 1:1 test cross in linkage.

Video Solution:



Q6 Text Solution:

Number of phenotypes in polygenic traits = $2n + 1$
 $5 = 2n + 1$ $n = 2$ gene pairs.

Video Solution:



Q7 Text Solution:

1 map unit = 1% recombination. So, 10% of gametes are recombinant.

Video Solution:



Q8 Text Solution:



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Height in humans is polygenic, not due to multiple alleles. Multiple alleles apply to traits like ABO blood group.

Video Solution:



Q9 Text Solution:

Chromosomes and their role in heredity were discovered after Mendel's work.

Video Solution:



Q10 Text Solution:

It is false — they often segregate together, not always. Crossing over and distance affect linkage.

Video Solution:



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